

HEW RESOURCE LIBRARY CODING GUIDE

TAB 1: REFERENCE INFORMATION

The Reference Information tab contains two coding fields (drop-down boxes) that need to be coded (notated by a *). The final field is an AI-generated “Study Objective” field. LaserAI creates this suggestion and it can be a good overview of the types of codes/topics/themes you may need to keep an eye out for while coding the article. Note: Take this with a grain of salt while we all get used to LaserAI. Please still read the article and code it yourself to verify any AI suggestions. Below are the two required coding fields:

1. **Reference Type:** What type of paper is it? **Reference Type** is a single value; you can choose only one.
 - **Research Article** (Article in peer-reviewed journal that involves original research. Reporting on the results of one or more studies, experiments, or models, and written by the person(s) who conducted the research; must contain new data. Abstract should mention some form of data collection or original analysis and present results.)
 - **Review Article** (Article in peer-reviewed journal that does not involve original research but provides an overview or review of the topic OR of specific literature. This category will include both qualitative topic reviews AND systematic or formal literature reviews. The latter should include a methodology for article inclusion and/or a search. Systematic reviews often include a meta-analysis which is used to synthesize the findings into a single quantitative summary. This category also includes the protocols published describing a planned systematic or scoping review.)
 - **Commentary/Opinion** (Article that puts forward an individual’s or organization’s viewpoint. Article in peer-reviewed journal that is labeled as either a Commentary, Editorial, Letter to the Editor, or other type of Opinion [e.g. Perspectives]. Coding will not include author responses or other responses to letters to the editor.)
 - **Assessment/Book/Report** (This reference category includes all gray literature reports and assessments from governmental and non-governmental organizations, as well as all items classified in their bibliographic information as “books.” Assessments [including subsections] that are published in peer-reviewed journals should be coded as Review Articles, not as Assessment/Book/Report.)
2. **Information Source:** What did the coder use as the source of their coding? Information source is a single value; you can choose only one.
 - **Complete Resource**
 - **Abstract and Title, Only**

TAB 2: EXPOSURE

Exposure pathway by which extreme weather affects health. Please note:

1. The Exposure field is a multi-value field; select all codes that apply, keeping in mind the coding multiple codes/category rule of 3.
2. The Exposure field does not need to be coded if it does not apply, however, please select the **Not Reported** option.

3. There are Level 2 coding values for 7 codes: Air Pollution, Extreme Weather-Related Event or Disaster, Food Quality, Food Security, Human Conflict/Violence, Temperature, and Water Quality.
4. Always include a Level 2 code if available. It is okay to only assign a first level code if no second level codes apply.
5. If an article briefly mentions an exposure pathway through which extreme weather can act on health, such exposure should be coded **ONLY** if it was discussed beyond merely framing a more general discussion.

Exposure Pathway Codes

- **General Exposure** (Use this code when there are more than 3 specific and unique exposures to code and none are discussed in depth.)

- **Air Pollution** (Apply Air Pollution codes when an air pollutant directly harms human health (e.g., impact cardiovascular health or asthma.)

- **Allergens** (Aero-allergens are airborne particles that cause allergic reactions in sensitized persons. Examples of ambient allergenic pollen include plant fungal spores, weed pollen [e.g., ragweed (*Ambrosia*)], grass pollen, and tree pollen.)
- **Dust** (This includes desert dust, dust emissions, and sand-dust weather.)
- **Ground-Level Ozone** (Ozone [O₃] is a gas and occurs both in the Earth's upper atmosphere and at ground level. Ground level ozone is a harmful air pollutant and is the main component of smog. Use this code in the context of the effect of weather on the health effects of **ground level ozone**. **Do not code in the context of how ozone as a greenhouse gas contributes to global warming or climate change.**)
- **Interaction with Temperature** (Health impact is mediated through the effects of an interaction between temperature and air pollution. An example would be a paper that concluded that interactions between high temperatures and wildfire air pollution should be considered in risk assessments regarding health consequences.)
- **Particulate Matter** (includes PM₁₀, PM_{2.5}, and black carbon)
- **Wildfire Smoke**
- **Other Air Pollution, Specify**

If a paper discusses air pollution in general or does not specifically discuss the listed pollutants such as dust or wildfire smoke, leave as a first level code (i.e., do not code a L2 option).

If **Wildfire Smoke** is selected, you should also code for **Particulate Matter** as wildfires are a major source of particulate matter.

- **Ecosystem Change** (An ecosystem refers to the animals, plants, and microorganisms that live in one place, as well as the environmental conditions that support them. An ecosystem change is any variation in the state, outputs, or structure of an ecosystem.

Ecosystem Changes should be used to code weather-related impacts on earth systems (i.e., wild ecosystems), **not man-made changes such as urbanization.**

Extreme weather can alter ecosystems (i.e., alter where species live, species diversity, lifecycles, and the timing of biological events) which can in turn impact human health. Ecosystem change can impact human health through its effect on food resources [e.g., fisheries], the health of agricultural systems, and the ecology of infectious disease organisms which result in altered distribution of disease vectors and pathogens.)

- **Extreme Weather-Related Event or Disaster** (Extreme Weather-Related Event includes unexpected, unusually severe, or unseasonal weather events; typically defined as occurring in the highest and lowest 5% of the historical distribution of similar weather parameters [though many publications will not have defined the term]. If an article discusses extreme weather in general, or mentions 4 or more specific types, only use the L1 code, Extreme Weather-Related Event/Weather Related Disaster.)
 - **Avalanche**
 - **Catastrophic Geologic Phenomenon** (i.e., glacial release, sinkhole)
 - **Dust Storm** (including haboob)
 - **Drought**
 - **Earthquake**
 - **Flood** (including riverine)
 - **Heatwave**
 - **Hurricane** (including cyclone and typhoon)
 - **Ice/Snow Storm** (including hail or extreme winter weather)
 - **Landslide** (including mudslide)
 - **Lightning Storm**
 - **Tornado**
 - **Tsunami**
 - **Volcanic Activity**
 - **Wildfire** (including variations on urban wildfires or similar terms)
 - **Windstorm** (including derecho)
 - **Other Extreme Weather-Related Event or Weather-Related Disaster, Specify**
- **Food Quality** (Attributes of human food, other than quantity, that affect health and well-being, e.g., biological or chemical contamination or changes in nutritional quality associated with extreme weather. Please note that chemical contamination of food [e.g., crops or fish/shellfish] is coded regardless of whether the chemicals were purposely applied (e.g., spraying of crops with pesticides) or introduced via flooding from an extreme weather event.)
 - **Crop/Plant Biotoxin** (Exposure to mycotoxins [e.g., aflatoxins] produced by certain fungi found on agricultural crops such as corn, peanuts, cottonseed, and tree nuts via consumption.)
 - **Crop/Plant Chemical** (Agricultural chemicals [e.g., pesticides, herbicides, insecticides, or fungicides] that are purposefully applied to crops, as well as other chemicals [e.g., dioxins, heavy metals, and hydrocarbons] that may be released and transported during extreme weather events such as hurricanes and floods and thereby contaminate crops.)
 - **Crop/Plant Pathogen** (Exposure to bacteria or viruses [e.g., E. coli or salmonella] via consumption of crops that have taken up contaminants through the soil or water.)

If **Heatwave** is selected, you should also code for **Temperature** -> **Extreme Heat/Heat**.

If **Ice/Snow Storm** is selected, you should also code for **Temperature** -> **Extreme Cold/Cold**.

The Disaster exposure codes are used when a *specific* heatwave or ice/snow storm event are being discussed. Otherwise, please use the Temperature exposure codes.

- **Livestock/Game Biotoxin** (Toxin produced by living organisms [e.g., Clostridium botulinum, mycotoxin] that contaminates foodstuffs from livestock or wild game [e.g., beef, pork, venison, poultry].)
 - **Livestock/Game Chemical** (Chemical [e.g., dioxin, veterinary pharmaceuticals] in the environment or purposely administered that contaminates foodstuffs from livestock or wild game.)
 - **Livestock/Game Pathogen** (Exposure to bacteria or viruses [e.g., E. coli or salmonella] via consumption of meat such as beef, pork, or poultry.)
 - **Marine/Freshwater Biotoxin** (Exposure to marine or freshwater biotoxins produced by certain types of toxic algae [e.g., harmful algal blooms] via consumption of fish and shellfish.)
 - **Marine/Freshwater Chemical** (Exposure to chemicals that accumulate in marine animals [e.g., heavy metals, PCBs, or organochlorine pesticides] via consumption of fish or seafood.)
 - **Marine/Freshwater Pathogen** (Exposure to bacteria in shellfish [e.g., *Vibrio parahaemolyticus* or *Vibrio vulnificus*] via consumption of seafood.)
 - **Nutritional Quality** (Includes the levels of essential nutrients [e.g., iron, zinc, or protein] in crops, livestock, or seafood. For example, higher water temperature may reduce the levels of iron and zinc in wheat, rice, maize, and soybeans.)
 - **Other Food Quality, Specify**
- **Food Security** (Impacts on sufficient physical and economic access to, and quantity of, food to meet dietary needs for an active and healthy life. Food access also includes issues of distribution, which may be impacted by extreme weather events/disasters.)

Food safety and nutrition should be coded as **Food Quality**.

 - **Availability/Distribution** (Availability of food can be affected by factors such as cost, quantity, and distribution. Distribution refers to the movement of food through supply chains and systems from producer to consumer [e.g., transport and distribution infrastructure].)
 - **Crop/Plant Food Security** (Aspects pertaining to agricultural or gathered plant foodstuffs and staples [e.g., wheat, barley, corn, cassava, wild/foraged berries].)
 - **Livestock/Game Food Security** (Aspects pertaining to farmed and wild terrestrial animal food sources [e.g., cattle, goats, poultry, hogs, deer rabbits, wild turkeys; includes milk and eggs].)
 - **Marine/Freshwater Food Security** (Aspects pertaining to marine or freshwater food projects [e.g., fish, shellfish, whale, seal]. Products can be obtained through capture/catch or aquaculture.)
 - **Other Food Security, Specify**
 - **Glacier/Snow Melt** (A perennial mass of ice, and possibly firn and snow, originating on the land surface by the recrystallisation of snow and showing evidence of past or present flow. A glacier typically gains mass by accumulation of snow, and loses mass by melting and ice discharge into the sea or a lake if the glacier terminates in a body of water. Land ice masses of continental size are referred to as ice sheets.)
 - **Green space/Blue space** (An area of grass, trees, or other vegetation, often set apart for recreational or aesthetic purposes in otherwise urban environments. These areas may

include gardens, parks, forests, or other natural areas. Blue space includes bodies of water or rivers, lakes, streams, ponds, pools, and fountains, often in conjunction with green space.)

- **Human Conflict/Violence**
 - **Human Conflict** (Conflict/Violence among groups/populations arising from extreme weather-induced or exacerbated events [e.g., conflicts over natural resources, forced migration, etc.]
 - **Interpersonal Violence** (Including gender-based, family/child, intimate partner. Violence between individuals resulting from extreme weather-induced psychological stressors such as heat and disasters and associated challenges, [e.g., power outages, evacuation, etc.]
 - **Other Human Conflict/Violence, Specify**
- **Indoor Environment** (Physical, biological, or chemical changes to the indoor environment related to extreme weather [e.g., **indoor temperature changes, mold resulting from flooding, changes to indoor air quality** associated with building ventilation adaptations] that lead to secondary health effects [e.g., asthma, heat stress, neurological impacts].)
- **Meteorological Factor** (Assesses health impacts associated with weather variables such as humidity, atmospheric pressure, or wind speed/direction.)
- **Precipitation** (Patterns of precipitation e.g., rainfall, snow, ice, include the amount, distribution, and timing of precipitation.)
- **Sea Level Rise** (Sea level rise is an increase in the volume of the water in the world's oceans, resulting in an increase in the global mean sea level.)
- **Sea Surface Oscillation** (A naturally occurring phenomenon that involves fluctuating ocean temperatures, including El Nino, La Nina, and Indian Ocean Dipole. This code should only be used when the pattern is the focus of discussion.)
- **Seasonality** (The periodic surge in disease occurrence corresponding to seasons or other calendar periods [e.g., increase in summer and decrease in winter]. Day length or the number of hours exposed to daylight should also be coded as Seasonality. Weather conditions [e.g., Precipitation, Temperature, or Solar Radiation] are often strongly associated with disease seasonality but are coded separately.)
- **Solar Radiation** (Solar ultraviolet radiation [UV/UVR; including UVA and UVB; i.e. sun exposure] is part of the electromagnetic spectrum emitted by the sun. Ozone depletion increases the amount of UV radiation that humans are exposed to. Only code when the paper looks at UV radiation or sun exposure. If paper looks at number of hours exposed to daylight, code **Seasonality**.)

Meteorological Factors do not include **Precipitation** and **Temperature**, which are **coded separately**. Specific meteorological factors (e.g., humidity) do not need to be specified as a Write-in.

- **Temperature** (Includes extreme cold, extreme heat, and temperature fluctuations.)
 - **Extreme Cold/Cold** (Temperature below an optimal level for health. Extreme cold/cold varies in different geographic regions depending on acclimatization. Wind chill factor is a measure of cold and describes what the air temperature feels like to the human skin due to the combination of cold temperatures and wind blowing on exposed skin.)
 - **Extreme Heat/Heat** (Temperature above an optimal level for health. Extreme heat/heat varies in different geographic regions depending on acclimatization. Heat index is a measure of how hot it really feels when relative humidity is factored in with the actual air temperature.)
 - **Variability** (Changes or shifts over short periods of time, e.g., days or less.)
- **Water Quality** (Biological or chemical contamination of water used for drinking, sanitation, or recreational water that is mediated or exacerbated by weather phenomena (e.g., temperature and precipitation). Please note that chemical contamination of water is coded regardless of whether the chemicals were purposely applied [e.g., spraying of crops with pesticides] or accidentally introduced via flooding from an extreme weather event.)
 - **Marine/Freshwater Biotoxin** (Exposure to biotoxins produced by certain types of toxic algae [e.g., harmful algal blooms] that may cause health impacts when inhaled, ingested, or absorbed.)
 - **Marine/Freshwater Chemical** (Chemicals, including agricultural chemicals [e.g., pesticides, herbicides, insecticides, and fungicides], synthetic or natural fertilizers [e.g., nitrogen and phosphorous], and other agents [e.g., dioxins, heavy metals, and hydrocarbons] that may be released and transported to contaminate water.)
 - **Marine/Freshwater Pathogen** (Exposure to waterborne pathogens including bacteria [e.g., *V. cholera*], viruses [e.g., norovirus], and parasitic protozoa [e.g., *Cryptosporidium*].)
 - **Other Water Quality, Specify**
- **Water Security** (Physical, social, and economic access to sufficient quantities of safe water for sustaining human consumption and sanitation. Impacts on water safety are tagged under Water Quality.)
- **Other Exposure, Specify**

If paper does not specify cold/decrease in temp or heat/increase in temp, or change in temp (variable), then just code Level 1 Temperature.

Articles discussing **Urban Heat Islands** are tagged under Temperature-> Heat and under Geographic Feature -> Urban

Articles that discuss **altitude** as a variable can be written in as Other Exposure, Specify –Altitude. Do not code as a Geographic Feature.

TAB 3: HEALTH IMPACT

Specification of health impact(s) related to extreme weather exposure.

Please note:

1. The Health Impact field is a multi-value field; select all codes that apply.
2. The Health Impact field does not need to be coded if it does not apply, however, please select the **Not Reported option**.

3. There are Level 2 coding values for 6 codes: Cardiovascular Impact, Developmental Impact, Infectious Disease, Mental Health/Well-Being, Respiratory Impact, and Temperature-Related Health Impact.
4. Always include a Level 2 code if available. It is okay to only assign a first level code if no second level codes apply. Within Infectious Disease, there are Level 3 codes available for Vectorborne Disease only (i.e., Mosquito-borne Disease). Otherwise, keep codes at Level 2.
5. Use of General ["Level 2 health impact"] code:
 - a. If a document covers **up to three** specific and unique concepts within a certain health impact category, then code for each of the concepts. If a document covers **more than three** specific and unique concepts, then use the secondary code General [health impact category] but **do not code** the individual concepts.
 - b. You can also use [General "Level 2 health impact" code] if the article either talks about category concepts only in general terms or does not specify a disease.

Health Impact Codes

- **General Health Impact** (Paper does not meet inclusion criteria for any specific health impact but generally mentions effect on health.)
- **Cancer**
- **Cardiovascular Impact**
 - **Heart Attack/Myocardial Infarction**
 - **Stroke**
 - **Hypertension/High Blood Pressure**
 - **Other Cardiovascular Impact, Specify**
- **Dermatological Impact** (Effects on the skin, nails, and hair)
- **Developmental Impact** (Impacts on human development prenatal to maturity)
 - **Birth Outcomes** (E.g., still birth, birth defects, low birth weight, fetal loss, small for gestational age, and preterm birth)
 - **Cognitive/Neurological/Psychological Disorder** (Disorders affecting development of general intellectual functioning such as reasoning, planning, judgment, abstract thinking, academic learning and experiential learning; specific developmental disorders of speech and language, motor function, pervasive developmental disorders)
 - **Pubertal Timing** (Changes in timing of onset of puberty)
 - **Stunting** (Impaired growth and development [e.g., low height-for-age])
 - **Other Developmental Impact, Specify**
- **Diabetes/Obesity/Overweight** (When coding, include Type 1 or Type 2 diabetes and changes in glucose intolerance and insulin sensitivity. Obesity can include outcomes such as overweight and body mass index.)
- **Infectious Disease**
 - **General Infectious Disease** (See Tab Instruction Note 5)

If a paper discusses cardiovascular disease in general and does not specifically discuss heart attack, stroke, or hypertension, leave as a first level code (i.e., do not code a L2 option).

- **Airborne Disease** (Diseases that are spread when droplets of pathogens are expelled into the air due to coughing, sneezing, or talking.)
 - **Foodborne Disease** (Foodborne diseases result from ingestion of food contaminated with microorganisms or chemicals. The contamination can occur at any stage in the food production-to-consumption pathway.)
 - **Vectorborne Disease** (Transmitted to humans by vectors e.g., fleas, flies, mosquitoes, and ticks.)
 - **Flea-borne Disease**
 - **Fly-borne Disease**
 - **Mosquito-borne Disease**
 - **Tick-borne Disease**
 - **Waterborne Disease** (Disease caused by drinking or bathing in water contaminated by pathogenic microorganisms such as bacteria, viruses, protozoa, and parasites.)
 - **Zoonotic Disease** (A disease transmissible from vertebrate animals to humans and vice-versa.)
 - **Other Infectious Disease, Specify** (infectious diseases that do not fit into any of the above categories [e.g., HIV])
- **Injury** (E.g., injury from extreme weather-related events or disasters, or secondary to other weather-related health impacts, e.g., fall due to heat stroke. Do not code visits to the emergency room; only code the specific health impact [e.g., Injury or Cardiovascular Impact].)
 - **Malnutrition** (Malnutrition arises from deficiencies of specific nutrients or from diets based on inappropriate combinations or proportions of foods, e.g., goiter, scurvy, anemia, and xerophthalmia are forms of malnutrition caused by inadequate iodine, vitamin C, iron, and vitamin A, respectively. Malnutrition includes both *undernutrition* and *overnutrition*. Undernutrition is insufficient intake of energy and nutrients to meet an individual's needs to maintain good health. Overnutrition is excessive intake of energy and nutrients beyond an individual's needs to maintain good health and can result in obesity.)
 - **Medical Visit** (E.g., hospital Admissions, emergency medical services—use if a paper presents the number of visits to a healthcare system (e.g., hospital, clinic, ER, ambulance dispatch) as a metric or surrogate for a health impact, especially if the paper is focused on metrics and economics. *If the paper presents the number of visits but specifies the health impact, then only code the health impacts and do not code for medical visit.*)
 - **Mental Health and Well-Being** (A person's emotional, psychological, and social well-being)
 - **General Mental Health and Well-Being** (Use this code for publications discussing general mental health, well-being, and quality of life. Includes stress, a state of mental tension that causes strong feelings of worry or anxiety, symptomatic mental disorders [e.g., dementia or delirium], mental, personality, and behavioral disorders

Suicide is coded under **General Mental Health and Well-being** and should also be tagged as an **Injury**.
Suicidal Ideation (i.e., suicidal thoughts and not suicide itself) has its own tag and is also coded under **Mood Disorder**.

due to brain damage, dysfunction, and physical disease, and suicide, which is also tagged as an injury.)

- **Childhood Behavioral Disorder** (Behavioral and emotional disorders with onset usually occurring in childhood and adolescence e.g., hyperkinetic disorders [attention-deficit/attention-deficit hyperactivity disorder (ADHD)], conduct disorders, mixed disorders of conduct and emotions, disorders of social functioning [e.g., autism spectrum disorder] and tic disorders.)
- **Mood Disorder** (Disorder primarily affecting emotional state [e.g., manic episode, bipolar affective disorder, depressive episode, recurrent depressive disorder, or persistent mood disorder].)
- **Schizophrenia/Delusional Disorder** (Disorder in which perception of reality is distorted [E.g., schizophrenia, schizotypal/schizoaffective disorder, persistent/induced delusional disorder, acute and transient psychotic disorder, other nonorganic psychotic disorder].)
- **Stress Disorder** (Disorder resulting from an atypical response to short- and long-term anxiety [E.g., phobic anxiety disorders, obsessive-compulsive disorder, reaction to severe stress, adjustment disorders, dissociative or conversion disorders, somatoform disorders, ecoanxiety, solastalgia, other anxiety or neurotic disorders].)

If a paper discusses stress disorders in general and does not specifically discuss ecoanxiety, PTSD, or solastalgia, leave as a first level code (i.e., do not code a L2 option).

- **Ecoanxiety** (The chronic fear of environmental cataclysm that comes from observing the seemingly irrevocable impact of extreme weather and the associated concern for one's future and that of next generations.)
- **PTSD, Post Traumatic Stress Disorder** (Disorder that develops when a person has experienced or witnessed a scary, shocking, terrifying, or dangerous event usually involving a situation where someone's life has been threatened or severe injury has occurred.)
- **Solastalgia** (Distress experienced by people due to environmental change that occurs while they are directly connected to their home environment [as opposed to nostalgia--the melancholia or homesickness experienced by individuals when separated from a loved home].)

- **Substance-Induced Disorder** (Mental and behavioral disorders due to psychoactive substance use [e.g., alcohol, opioids, cannabinoids, sedatives, cocaine, stimulants, hallucinogens, tobacco, VOCs].)
 - **Suicide Ideation** (Suicidal thoughts but not the act of suicide)
 - **Other Mental Disorder, Specify**
- **Morbidity/Mortality** (Morbidity is the condition of being diseased. Mortality is death, especially on a population scale. Use this code for papers that include rates of mortality or morbidity or the burden of disease from mortality and morbidity is quantified using metrics [e.g., Disability-Adjusted Life Year, or DALY].)
 - **Neurological Impact** (Diseases/disorders affecting the brain and central and autonomic nervous systems and result in symptoms including paralysis, muscle weakness, poor coordination, loss of sensation, seizures etc. [e.g., Alzheimer's Disease, epilepsy, multiple sclerosis, Parkinson's Disease, ALS, migraines].)

- **Reproductive Impact** (Effect on reproductive system or outcome such as infertility, sperm quality, pregnancy, puberty, or menopause)
- **Respiratory Impact**
 - **Asthma**
 - **Bronchitis/Pneumonia**
 - **Chronic Obstructive Pulmonary Disease (COPD)**
 - **Interstitial Lung Disease** (E.g., Sarcoidosis, idiopathic pulmonary fibrosis)
 - **Lung Cancer**
 - **Upper Respiratory Allergy** (E.g., Hay fever, seasonal allergies, or pollinosis)
 - **Other Respiratory Impact, Specify**
- **Temperature-Related Health Impact**
 - **Cold-Related Health Impact** (E.g., hypothermia)
 - **Heat-Related Health Impact** (E.g., hyperthermia, heat stress, heat stroke, heat exhaustion)
 - **Thermal Comfort**
- **Urologic Impact** (Effect on the renal system [i.e., urinary or kidney system], including acquired dysfunction or exacerbation of congenital conditions [e.g., kidney disease, kidney failure, kidney stones (i.e., nephrolithiasis)].)
- **Other Health Impact, Specify**

If a paper discusses respiratory impact in general and does not specifically mention specific impacts such as asthma, leave as a first level code.

For publications about antimicrobial or antibiotic resistance, coders should use the standardized term **Antimicrobial Resistance** as the write-in under Other Health Impact. Antibiotic resistance is a subset of the overall category which may be added as a code in the future.

TAB 4: GEOGRAPHY

Use the geographic location codes to tag the geographical area discussed in the reference. Code only for region, not specific countries. If a paper briefly mentions several regions, a specific region should be checked only if it was included beyond merely framing the more general discussion. **Select all that apply. If no specific country or region meets these criteria, select Global or Unspecified Location.** This field should always be coded.

If a paper discusses two or fewer geographical locations, then code each one separately. If a paper discusses three or more locations, then code as Global or Unspecified Location. If a paper is discussing the Arctic broadly and not a specific country or region within the Arctic, then code geographic location as Global or Unspecified Location and geographic feature as Polar.

Geography Codes

Geographic Location

- **Global or Unspecified Location**
- **Non-United States**

- **Africa** (Includes all countries within the continent of Africa, e.g., Algeria, Botswana, Egypt, Ethiopia, Ghana, Kenya, Madagascar, Mali, Morocco, Nigeria, South Africa, Togo, Uganda, Zambia, and Zimbabwe, etc.)
- **Antarctica**
- **Asia** (Asia the continent is bounded by Europe and the Arctic, Pacific, and the Indian oceans and includes:
 - East Asia [China, Mongolia, Hong Kong, Japan, Korea (North and South), Taiwan]
 - Southeast Asia Cambodia, Indonesia, Malaysia, Philippines, Thailand, Vietnam
 - Central Asia Countries including Afghanistan, Kazakhstan, Bangladesh, India, Iran, Pakistan, Sri Lanka
 - Guam/Mariana Islands [in Micronesia in the western Pacific Ocean; See table below for specific Pacific Islands]
 - Western Asia/Middle East [Armenia, Iraq, Israel, Jordan, Kuwait, Saudi Arabia, Syria, United Arab Emirates])
- **Australasia** (A region of Oceania, comprising Australia [including Tasmania], New Zealand, and neighboring islands in the Pacific Ocean including New Guinea, Papua New Guinea, American Samoa; located in the South Pacific Ocean; New Caledonia; See table below for specific Pacific Islands)
- **Central/South America**
 - Central America includes Guatemala, Belize, Honduras, Nicaragua, El Salvador, Costa Rica, and Panama.
 - South America includes Argentina, Bolivia, Brazil, Chile, Columbia, Ecuador, Paraguay, Peru, Uruguay, and Venezuela.
- **Europe** (France, Germany, Austria, Spain, Italy, Sweden, Denmark, Belgium, Norway, Finland, Poland, United Kingdom, Ireland, Romania, Belarus, Russia, Ukraine, Greece, Cyprus, Bulgaria, Hungary, Portugal, Serbia, Lithuania, Croatia, Turkey, Iceland)
- **Non-U.S. North America** (includes Canada, Greenland, Mexico, and Bermuda)
 - Also include in this region the Caribbean countries/territories of the Dominican Republic, Cuba, Haiti, Virgin Islands (non-U.S.), Jamaica, the Bahamas, Antigua & Barbuda, Trinidad, and Tobago.
- **United States**
 - Also include in this region U.S. territories of Puerto Rico and U.S. Virgin Islands

Pacific Islands/Territories/Nations	Current Coding Guide
Federated States of Micronesia (FSM)	Asia
Guam	Asia
Northern Mariana Islands (CNMI)	Asia
Palau	Asia
Marshall Islands	Australasia
American Samoa	Australasia
Cook Islands	Australasia

Fiji Islands	Australasia
French Polynesia	Australasia
Kiribati	Australasia
Nauru	Australasia
New Caledonia	Australasia
Niue	Australasia
Papua New Guinea (PNG)	Australasia
Pitcairn Islands	Australasia
Samoa	Australasia
Solomon Islands	Australasia
Tokelau	Australasia
Tonga	Australasia
Tuvalu	Australasia
Vanuatu	Australasia
Wallis and Futu	Australasia

Geographic Feature

Use these codes to tag specific environmental or geographical features **discussed in the paper**. Coded Geographic Features should be directly relevant to the **focus** of the paper. If a paper briefly mentions multiple environmental or geographical features, a specific feature should be checked only if it was included beyond merely framing the more general discussion. **Select all that apply. If a paper does not discuss geographic or environmental features, you do not need to apply code. Please select the Not Reported option.**

- **General Geographic Feature** (Paper generally mentions multiple features only in order to frame the discussion.)
- **Built Environment** (Manmade geographical features such as highways, bridges, airports, schools, hospitals)
- **Desert** (Arid land with usually sparse vegetation e.g., desert, semi-desert, arid, semi-arid, sub-Saharan, Sahel.)
- **Forest** (E.g., forest, woodlands, boreal, boreal forest, but NOT rainforest which has a separate code)
- **Freshwater** (E.g., freshwater lake, river, river basin, riverine)
- **Grassland** (E.g., grasslands, savanna, prairie, moorlands, lowlands)
- **Island** (A tract of land surrounded by water and smaller than a continent. Very small islands can be called islets, skerries, cays, or keys.)
- **Mountain** (E.g., alpine, highlands)
- **Ocean/Coastal** (Coastal refers to land near an ocean shore. Include articles that discuss coastal plains, beaches, or beachfront.)
- **Polar** (E.g., Arctic, sub-arctic, Antarctic, sub-Antarctic, glacier, tundra)
- **Rainforest** (Tropical woodland with an annual rainfall of at least 100 inches and marked by lofty broad-leaved evergreen trees forming a continuous canopy.)

If a paper is discussing the Arctic broadly and not a specific country or region within the Arctic, then code location as Global or Unspecified Location and feature as Polar.

- **Rural** (E.g., farmland, cropland)
- **Temperate** (Temperate climates are those without extremes of temperature and precipitation [rain or snow]; moderate climate or weather.)
- **Tropical** (Occurring in the tropics which is the region on either side of the equator. Sub-tropical refers to the geographic and climate zones located between the tropics and the temperate zones.)
- **Urban** (Relating to, characteristic of, or constituting a city e.g., urban, peri-urban, semi-urban, suburban.)
- **Valley** (A low area of land between hills or mountains, typically with a river or stream flowing through it.)
- **Wetland** (Areas where water covers the soil, or is present either at or near the surface of the soil all year or for varying periods of time during the year, including during the growing season e.g., marshland or swampland.)
- **Other Geographic Feature, Specify**

Specific regions (e.g., Mediterranean) are not considered a geographic feature and should not be written in. **Altitude** is also not considered a geographic feature and should be instead written in as an exposure.

TAB 5: DATA TOOLS & METHODS

Description: Papers requiring data tools and methods codes will utilize data resources as a core part of the methodology or design, build, or create a tool for future use in similar study contexts. Papers may include **data resources**, such as datasets, software code, or cohorts, as well as **models**, such as geospatial, exposure, or artificial intelligence models.

Please note:

1. The Data Tools & Methods field is a multi-value field; select all codes that apply.
2. The Data Tools & Methods field does not need to be coded if it does not apply, however, please select the **Not Reported** option.

Data Tools & Methods Codes

Data Resource Type (Specific type of data resource used in paper.) **Check all that apply.**

- **Cohort** (A cohort is a group of individuals with a shared characteristic observed over a period of time to examine how specific factors impact their health outcomes. Cohort studies are often used to investigate how exposures affect health outcomes over time.)
 - **Source Cohort Publication** (If a publication is establishing a **new** cohort rather than utilizing an existing cohort's data.)
- **Dataset** (Publications that mention depositing their data in a repository or otherwise making it available online for other researchers to access. Dataset should not be tagged if it is not open access/publicly available. Checking the data availability section at the end of an article is often helpful – if data is “available upon request,” dataset should **not** be tagged. **Example language:** “The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.”)
- **Software Code/Library** (Software code are instructions written in a programming language that tells a computer what to do. A software library is a collection of pre-developed software code that provides specific functions, allowing developers to reuse existing code for common tasks without having to develop them from scratch.)

- **Survey** (A method of gathering information from a sample of people using questions. Survey should **not** be tagged if it is not open access/publicly available or included within the publication.)
- **Other Data Resource Type, Specify**

Model Type (Specific type of model used in paper.) **Check all that apply.**

- **Artificial Intelligence/Machine learning** (Key terms may include artificial intelligence, machine learning, deep learning, neural networks, random forest, decision trees, backpropagation. Also look for mention of popular machine learning software such as "Scikit-learn," "TensorFlow," "Keras," "PyTorch," "XGBoost," "LightGBM".)
- **Exposure Modeling** (An exposure model predicts how a human is exposed to a specific environmental pollutant or chemical. The aim is to estimate the exposure over a specific time period considering how the substance moves through the environment and how individuals come into contact with it. They can be used for various purposes, including risk assessment and regulatory decisions.)
- **Geospatial Modeling** (A mathematical and digital representation of geographic phenomena. These models integrate various types of spatial data to represent geographic features and use computational approaches to simulate and analyze spatial patterns and processes. They are developed to address specific research questions such as understanding disease spread or predicting flood risks.)
- **Other Model Type, Specify**

TAB 6: SPECIAL TOPICS

Description: Special topics relevant to extreme weather impacts on health that may be a part or focus of a reference's discussion.

Please note:

3. The Special Topics field is a multi-value field; select all codes that apply.
4. The Special Topics field does not need to be coded if it does not apply, however, please select the **Not Reported** option.
5. There are Level 2 coding values for 4 codes: Communication, Intervention, Sociodemographic Vulnerability, and Study Population.
6. Always include a Level 2 code if available. It is okay to only assign a first level code if no second level codes apply.

Special Topics Codes

- **Climate Justice/Climate Equity** (Climate Justice refers to a view of climate change as an ethical issue and consideration of the relationship of its causes and effects to concepts of environmental and social justice, equality, human rights, collective rights, and historical responsibility. **Climate Equity** refers to a view of climate change as an ethical issue that considers the relationship of its causes and effects to concepts of environmental and social justice, equality, human rights, collective rights, and historical responsibility.)
- **Communication** (Papers include research, methods, theoretical papers on how to communicate or frame various issues related to extreme weather and human health as well as surveys of attitudes,

perceptions, knowledge, and beliefs related to extreme weather. Typically, an audience is explicitly mentioned or implied.) **Select all that apply.**

- **General Public/Unspecified** (If audience is unspecified, it is presumed to be General Public.)
 - **Community/Disease Advocacy/ Non-Governmental** (Specific organized groups or individuals engaged in encouraging policy or other actions in a public manner [e.g., advocate, funder, supporter, member])
 - **Educator/Student** (Educator or student at any level. Researchers are tagged separately. Teachers or students who are not researchers, curricula developers, educational advocacy organizations)
 - **Health Professional** (Professional working in the clinical health or public health field e.g., physicians, nurses, public health professionals, community health workers)
 - **Policymaker** (E.g., government officials, legislators, members of international bodies, legal professionals)
 - **Researcher** (E.g., academic, government, industry, or private individuals engaged in research; includes all scientific disciplines)
 - **Other Communication Audience, Specify**
-
- **Economic Impact** (Papers include discussion of the economic costs of extreme weather health impacts. This could include health-related costs of disasters [historical or estimated], modeled predictions of economic costs, or reviews of other studies of economic costs. Papers can also focus on estimates of costs avoided or saved by actions to adapt to extreme weather.)
 - **Health Sector Influence** (The health care sector, including clinical health care and public health systems is engaged in adaptation to extreme weather impacts. Papers could address topics such as the role of the health sector [including health professionals] in advocating for extreme weather adaptation strategies or resilience and other adaptation measures for health care systems including non-human components of health care systems [e.g., buildings, energy and water sources, health campuses] as well as human components [e.g., resilience of health care facility staff]. Papers that focus only on public health or individual clinical interventions for specific health impacts should NOT be included.)
 - **Intervention** (Extreme weather interventions are a focus of the paper or a significant part of analysis or discussion. Interventions are a specific adaptation or disaster planning/preparedness action that attempts to reduce the impact of extreme weather on health. This could include changes in policy or infrastructure that allow adaptation to extreme weather. Papers could describe proposed interventions, an evaluation of an intervention, or methods for prioritizing interventions.)
 - **Disaster Risk Reduction** (References specifically refer to the term “Disaster Risk Reduction” as a focus. Papers may also refer to the “Sendai Framework” which provides a policy guidance for disaster risk reduction.)
 - **Early Warning System** (An early warning system is used to warn a population of high temperatures, hurricanes, and other hazards.)
 - **Vulnerability Assessment** (Vulnerability assessments are conducted to identify the people at highest risk of effects from extreme weather **within a particular location or community**. Vulnerability assessments are undertaken in the context of efforts to protect vulnerable populations and take into account not just exposure and sensitivity of a population, but also its capacity to adapt to or reduce its risk. In contrast, risk

assessments are used to characterize risks for other purposes, such as policy analysis. The *terminology* used in the paper (i.e., “vulnerability”, “risk,” or “impact” assessment), however, is less important than the *context* of the study and its **inclusion of adaptive capacity**. A study that does not include adaptive capacity should not be coded. **Papers that describe methods for conducting vulnerability assessments without specific case studies or analyses should be in**

- **Policy** (The focus or substantive portion of the reference is a policy analysis, describes a policy agenda, or makes specific policy recommendations related to extreme weather and health impacts (e.g., policies aimed at application of research or implementation of measures) or discussed in some detail the need for specific policy actions.)
- **Population Displacement/Forced Migration** (Extreme weather-induced movement/relocation of people away from their home or home region to another geographic area [e.g., due to drought-induced famine]. Forced migration is involuntary and may be due to fear of harm from conflict or natural disasters, or because of political decisions.)
- **Research Gap** (Description or analysis of gap in the knowledge base on the impacts of extreme weather on human health with a level of specificity beyond a statement of need for further research. Articles can be coded for Research Gaps that are described for any of the Tabs or codes under them e.g., could be about research gaps on exposures, impacts, etc.)
- **Sociodemographic Vulnerability** (Papers focus on sociodemographic factors such as income, education, wealth, and empowerment that may affect an individual’s vulnerability to extreme weather health impacts.)

If Sociodemographic Vulnerability is coded, it is likely that Study Population -- Low Socioeconomic Status, will also apply.

 - **Healthcare Access** (A lack of access to health care coverage and services in a timely manner from qualified providers can affect an individual’s or population’s health outcomes. This can include care from hospitals as well as other sources such as providers, public health clinics, etc. This code should be used when the lack of access is an existing, chronic problem impacting a population’s vulnerability to extreme weather health impacts.)
- **Study Population** (Papers with an **emphasis on a specific population and extreme weather health impacts**. Papers can include those in which analysis is conducted/reported **wholly** within a particular population or for subgroups of populations i.e., if the group is discussed in depth in the discussion section.) **Select all that apply.**

- **General Population** (Paper does not specify individual populations but refers to general population.)
- **Athletes/Recreational Warriors**
- **Children**
- **Displaced Populations** (populations displaced directly or indirectly by extreme weather)
- **Elderly**
- **Farmers** (I.e., a person who owns or manages a farm)
- **Sex** (I.e., subpopulations who experience impacts directly or disparately as a result of their sex)
- **Incarcerated Persons**
- **Indigenous People:** Communities that live within, or are attached to, geographically distinct traditional habitats or ancestral territories, and who identify themselves as being part of a distinct cultural group which maintains a cultural and social identity distinct from the mainstream or dominant culture. Add specific groups as write ins e.g., Inuit, Yupik, Yanonmani, etc.
- **Low Socioeconomic Status** (I.e., impoverished)
- **People with Disabilities**
- **Pre-existing Medical Condition** (i.e., persons who have a pre-existing or underlying medical condition that is directly affected or exacerbated by one or more exposures.)
- **Pregnant or Breastfeeding Women**
- **Unhoused Persons/People Experiencing Homelessness**
- **Workers**
- **Indigenous/Racial/Ethnic Subgroup, Specify**
- **Other Study Population, Specify**

There are some groups that may be added as new terms in the future. Coders should use these standardized terms for write-ins:

- **Chronic Mental Condition**

In general, do not add a specific occupation as a Write-in (e.g., Firefighters) when coding for Workers.

When coding for Farmers, also code for Workers.